A CRITERION FOR K-INJECTIVITY IN A GROTHENDIECK ABELIAN CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, each of the successor and limit steps being written out; the only inputs it names are same-chapter and cross-chapter results carried on this card as navigation stubs, and no step is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Acyclic complexes of bounded size inside an acyclic complex. Same chapter, Lemma lemma-acyclic-quotient-complexes-bounded-size. Supplies, for a nonzero acyclic quotient, a subcomplex of the cardinal-bounded size that the transfinite construction inserts at successor steps.
- The complex of morphisms and its cohomology. More on Algebra, Section more-algebra-section-hom-complexes and Equation more-algebra-equation-cohomology-hom-complex. Supply the identification of the complex of morphisms with the groups computing morphisms in the homotopy category and the limit description at a limit ordinal.
- Mittag-Leffler for ordinals. Homology, Lemma homology-lemma-ML-over-ordinals. Supplies the vanishing of the limit term at a limit ordinal, which closes the transfinite induction.
- K-injective complexes. Same chapter, the definition of a K-injective complex. Supplies the property that the proof establishes, namely the vanishing of morphisms from every acyclic complex.

Source reference. Original lemma 079K: sources/stacks-injectives/pinned-original-source.tex, source lines 1808–1808. Navigation pointer invoked at proof line 1853; the counted span contains the pointer only and the cited result is not reproduced

Date: stacks-project:injectives.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 0A8H: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17851–17851. Navigation pointer invoked at proof line 1909; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original equation 0A5X: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17875–17875. Navigation pointer invoked at proof line 1912; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0AAT: sources/stacks-homology/pinned-original-source.tex, source lines 8079–8079. Navigation pointer invoked at proof line 1917; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. Let κ be a cardinal as in Lemma 079K. Suppose that $I^\bullet$ is a complex such that*

1. *each I^j is injective, and*
2. *for every bounded above acyclic complex $M^\bullet$ such that $|M^n| \leq \kappa$ we have $\mathrm{Hom}_{K(\mathcal{A})}(M^\bullet, I^\bullet) = 0$.*

Then $I^\bullet$ is an K-injective complex.

Proof. Let $M^\bullet$ be an acyclic complex. We are going to construct by induction on the ordinal α an acyclic subcomplex $K^\bullet_\alpha \subset M^\bullet$ as follows. For $\alpha = 0$ we set $K^\bullet_0 = 0$. For $\alpha > 0$ we proceed as follows:

1. If $\alpha = \beta + 1$ and $K^\bullet_\beta = M^\bullet$ then we choose $K^\bullet_\alpha = K^\bullet_\beta$.
2. If $\alpha = \beta + 1$ and $K^\bullet_\beta \neq M^\bullet$ then $M^\bullet/K^\bullet_\beta$ is a nonzero acyclic complex. We choose a subcomplex $N^\bullet_\alpha \subset M^\bullet/K^\bullet_\beta$ as in Lemma 079K. Finally, we let $K^\bullet_\alpha \subset M^\bullet$ be the inverse image of $N^\bullet_\alpha$.
3. If α is a limit ordinal we set $K^\bullet_\beta = \mathrm{colim}\, K^\bullet_\alpha$.

It is clear that $M^\bullet = K^\bullet_\alpha$ for a suitably large ordinal α. We will prove that

$$\mathrm{Hom}_{K(\mathcal{A})}(K^\bullet_\alpha, I^\bullet)$$

is zero by transfinite induction on α. It holds for $\alpha = 0$ since $K^\bullet_0$ is zero. Suppose it holds for β and $\alpha = \beta + 1$. In case (1) of the list above the result is clear. In case (2) there is a short exact sequence of complexes

$$0 \to K^\bullet_\beta \to K^\bullet_\alpha \to N^\bullet_\alpha \to 0$$

Since each component of $I^\bullet$ is injective we see that we obtain an exact sequence

$$\mathrm{Hom}_{K(\mathcal{A})}(K^\bullet_\beta, I^\bullet) \leftarrow \mathrm{Hom}_{K(\mathcal{A})}(K^\bullet_\alpha, I^\bullet) \leftarrow \mathrm{Hom}_{K(\mathcal{A})}(N^\bullet_\alpha, I^\bullet)$$

By induction the term on the left is zero and by assumption on $I^\bullet$ the term on the right is zero. Thus the middle group is zero too. Finally, suppose that α is a limit ordinal. Then we see that

$$\mathrm{Hom}^\bullet(K^\bullet_\alpha, I^\bullet) = \lim\nolimits_{\beta<\alpha} \mathrm{Hom}^\bullet(K^\bullet_\beta, I^\bullet)$$

with notation as in More on Algebra, Section 0A8H. These complexes compute morphisms in $K(\mathcal{A})$ by More on Algebra, Equation (0A5X). Note that the transition maps in the system are surjective because I^j is injective for each j. Moreover,

for a limit ordinal α we have equality of limit and value (see displayed formula above). Thus we may apply Homology, Lemma 0AAT to conclude. $\square$